

CE 560: Soil Dynamics

Course Number: CE 560

Course Name: Soil Dynamics

L-T-P-C: 3-0-0-3

Prerequisites: Geotechnical Engineering I (CE 302), Geotechnical Engineering II (CE 402) or Equivalent course.

Intended for: UG and PG students

Distribution: Discipline Elective

Approval: 28th Senate

Course Contents

- **Introduction to soil dynamics:** Importance of soil dynamics, nature and types of dynamic loading, concept of dynamic loading, Simple harmonic motion, degrees of freedom, types of vibration, free and forced vibration, decay of motion, vibration measuring instruments. [8 Lectures]
- **Wave propagation in elastic half space:** Elastic response of continua, Wave propagation in soil media, Earthquake waves, Seismic travel time curve, Three-circle method of estimation of Earthquake epicentre. [6 Lectures]
- **Dynamic soil properties:** Stresses in soil element, concept of soil stiffness, damping ratio and plasticity properties of soil, techniques for estimation of dynamic soil properties from field (intrusive and non-intrusive testing) and laboratory testing. Correlation for obtaining various geotechnical parameters. [6 Lectures]
- **Site response analysis:** Transfer function, homogeneous and layered un-damped and damped soil resting on rigid block, equivalent and non-linear approach of ground response analysis, convolution and deconvolution of the earthquake wave, site classification. [6 Lectures]
- **Liquefaction and lateral spreading:** Concept of liquefaction (static and cyclic), cyclic mobility, estimation of liquefaction potential of soil from field and laboratory testing data, cyclic stress ratio of soil and its importance, post liquefaction lateral spreading of soil, effect of liquefaction on various geotechnical structures. [6 Lectures]
- **Design of machine foundations:** Types of machine foundation, Mass-spring dashpot model, concept of vibration isolation, Tschebotarioff's reduced natural frequency method; Elastic half space method; Vertical, sliding, torsional (yawing) and rocking (and pitching) modes of oscillations; Design guidelines as per codes; Typical design problems. [10 Lectures]

Textbooks:

1. Shamsheer Prakash, **Soil Dynamics**, McGraw Hill Book Company, 1981.
2. Steven L. Kramer, **Geotechnical Earthquake Engineering**, Prentice Hall, 2003.

References:

1. Robert W. Day, **Geotechnical Earthquake Engineering Handbook**, McGraw Hill, 2002.

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2. Kenji Ishihara, **Soil Behaviour in Earthquake Geotechnics**, Oxford University Press, 1996.
 3. G.V. Ramanna and B.M. Das, **Principles of Soil Dynamics**, CENGAGE Learning, 2011.
 4. Richart, F.E., Woods, R.D. and Hall, J.R., **Vibrations of soils and foundations**, Prentice-Hall, 1970.

Additional Readings

1. Journal papers in the area of Soil Dynamics.